SUPPORTING ADULT GUIDANCE

A line as a relationship

Straight Lines and Connected Representations. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner connects the five representations of a linear relationship — context, table, coordinates, equation and graph — and meets the equation y **= mx + c**. The emphasis is on **fluency of translation**: seeing that these are one relationship, and that its two numbers are the y-intercept (*c*, the value of y when $x = 0$) and the constant step (*m*). Formal work on gradient and intercept follows in later lessons; here the goal is the connected picture.

Driving Question: *Can a line tell the same story as an equation, a table and a graph?*

The mathematics, in plain terms

A **linear relationship** is one whose graph is a straight line: every time x increases by 1, y changes by the same fixed amount. It can be written as an equation, y **= mx + c**. Here **c** is the y**-intercept** — the value of y when $x = 0$, which is where the line crosses the y-axis (a line has no real 'start'; in a context where x counts from 0, this is the initial value). **m** is the **step** — how much y changes each time x goes up by 1 (later called the gradient). The same relationship can be told as a **story** (“£4 to join, then £2 a visit”), listed as a **table**, plotted as **coordinates**, written as an **equation** ($y = 2x + 4$), or drawn as a **graph**. None of these is more “real” than the others; they are five views of one thing. The power this lesson builds is moving between them — and the key that unlocks every move is finding the two numbers: where it crosses the y-axis, and how much it steps.

One relationship, five views

The heart of this lesson is a shift from seeing tables, graphs and equations as separate topics to seeing them as one relationship in different costumes. The most useful thing to protect is that the learner drives the sliders and watches everything move together — that simultaneous change is the insight, and it cannot be got from a static worked example. Keep the two numbers in plain language at first: the *value when* $x = 0$ and the *step*. The formal names, gradient and intercept, arrive in the next few lessons; introducing them too hard, too early can obscure the simple idea. This lesson is the spine of the whole unit — every later lesson is a closer look at *m* or *c*, so time spent making the connected picture secure pays back many times over.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	One Line, Five Costumes: drag m and c and watch the story, table, coordinates, equation and graph change together.	~15 min

Movement	Step the Story: move along the line one x at a time and see the same step repeat.	~8 min
Reflection	Match the Representation: read a table or story and choose the matching equation.	~10 min
Creativity	Model the Story: set m and c so a line fits a real situation.	~5 min
Rest	A pause after holding one relationship in five forms at once.	~4 min
Nutrition	Contextual (Mode A): tariffs, fares, wages and conversions — a fixed start plus a steady rate.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking the table, graph and equation are separate, unrelated things

Repair: “Change m on the slider. Which of the other views changed? So are they separate, or one relationship?”

Mixing up m and c

Repair: “Which number is the value when $x = 0$ (the y-intercept), and which tells you how much it steps each time?”

Reading a coordinate as (up, across) instead of (across, up)

Repair: “Which number comes first in a coordinate pair — the across (x) or the up (y)?”

Assuming a line must go up (positive step)

Repair: “What happens to the line when m is negative? Does y go up or down as x increases?”

Thinking c is always where the line is highest or lowest

Repair: “c is simply the value when $x = 0$ — where the line meets the y-axis. Is that always the top or bottom?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: reading and plotting coordinates, substituting a value into a simple rule, and reading a table of values. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the two numbers — and the translation — with the learner:

- “What is the value when x = 0? That is c, the y-intercept.”
- “How much does it change each time x goes up by 1? That is m, the step.”
- “Can you write that as y = mx + c?”
- “Now show the same line as a table and as a graph.”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Should they learn the words gradient and intercept now?”
They will meet both formally in the next two lessons. Here, ‘step’ and ‘the value when x = 0’ are enough — the priority is the connected picture, not the vocabulary.

“Why start with all five representations at once?”
Because the whole point is that they are one relationship. A learner who meets tables, graphs and equations separately often never connects them; seeing them move together fixes that from the start.

“They find negative steps confusing.”
Let them drag m below zero and watch the line fall, then tell a story for it (“a tank losing a litre a minute”). Negative gradients get a dedicated reconnection session soon; a first, gentle encounter here is enough.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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